Food Delivery Business Performance & Market Analytics Report

Introduction

The Food Delivery Performance & Market Analytics project was developed to analyse food delivery orders, customer behaviour, revenue performance, restaurant categories, delivery efficiency, and city wise market performance using data analytics techniques. The project integrates Microsoft Excel, Python, MySQL, and Power BI to transform raw food delivery data into meaningful business insights. The interactive dashboard helps identify high performing cities, restaurant categories, revenue trends, customer satisfaction levels, and delivery time patterns, enabling businesses to make informed and data driven decisions.

Objectives

1. To analyse food delivery order and customer data using data analytics techniques.

2. To evaluate total revenue and average order value.

3. To identify high performing and low performing cities based on revenue.

4. To analyse restaurant category wise revenue and performance.

5. To analyse customer satisfaction using customer ratings.

6. To evaluate delivery time performance and its relationship with customer ratings.

7. To identify high value customers based on their total spending.

8. To analyse monthly revenue trends and identify changes in business performance.

9. To develop an interactive Power BI dashboard for business analysis and decision making.

Tools Used

1 Microsoft Excel – Raw Data Inspection and Initial Analysis

Used to inspect the raw dataset, identify rows and columns, check data types, missing values, and duplicate records.

2 Python – Data Cleaning and Preparation

Used to remove duplicate records, handle missing and invalid values, standardise city and payment method values, clean currency formats, convert data types, and prepare the cleaned dataset for further analysis.

3 MySQL Workbench – Database Management and SQL Analysis

Used to store the cleaned dataset and perform SQL based business analysis such as total revenue, customer count, average order value, city wise revenue, category performance, monthly revenue, and high value customer analysis.

4 Microsoft Power BI – Data Visualisation and Dashboard Development

Used to create KPI cards, monthly revenue trends, city wise revenue visualisations, restaurant category performance charts, and delivery time versus customer rating analysis through an interactive dashboard.

Dashboard Overview

The Power BI dashboard presents key food delivery business insights through KPI cards and interactive visualisations. It includes Total Revenue, Total Customers, Average Customer Rating, Monthly Revenue Trend, City wise Revenue, Restaurant category Performance, and Delivery Time versus Customer Rating analysis. The dashboard enables users to compare city and restaurant category performance, monitor revenue trends, evaluate customer satisfaction, and identify areas requiring business improvement. These insights support better marketing, restaurant partnership, delivery management, and customer focused business decisions.

Key Findings

1. Total Revenue

The analysis recorded a total revenue of ₹117,420, representing the overall order value generated during the analysed period.

2. Customer Rating and Delivery Performance

The average customer rating was 4.04, indicating a generally positive level of customer satisfaction. However, further improvement can be achieved by reducing delivery times and improving delivery efficiency, which may contribute to higher customer satisfaction and ratings.

3. Best Performing City

Kochi was identified as the best performing city, generating the highest total revenue of ₹24,130 among the analysed cities.

4. Restaurant Category Performance

Biryani was the strongest performing restaurant category, generating total revenue of ₹40,200. In contrast, Café was the weakest performing category, with total revenue of ₹7,620, indicating an opportunity for targeted promotional and growth strategies.

5. Peak Monthly Revenue

April recorded the highest monthly revenue, with a total revenue of ₹32,040, making it the peak performing month during the analysed period.

Recommendations

1. Improve Performance in the Kollam Market

• Focus on improving Kollam, which records the lowest revenue among the cities.

• Expand restaurant availability in Indian, Fast Food, Chinese, Desserts, and Café categories.

• Introduce targeted offers, discounts, and coupons to increase customer engagement.

• Improve delivery speed and restaurant partnerships to enhance customer satisfaction.

• Maintain Biryani's strong performance while promoting other food categories to drive balanced growth.

2. Improve Customer Satisfaction and Category Performance in Calicut

• Improve delivery speed to increase customer satisfaction, as Calicut has a relatively lower average rating of 3.86.

• Strengthen Chinese and Desserts categories through better restaurant availability and targeted offers.

• Maintain the strong performance of Biryani while promoting other categories for balanced growth.

• Address the May revenue decline through seasonal promotions and suitable food recommendations, particularly refreshing and lighter options during warmer months.

3. Introduce Seasonal and Trending Food Promotions

• Address the May revenue decline through targeted seasonal food promotions.

• Promote seasonal and refreshing food options suited to warmer months.

• Introduce trending and social media popular food items through restaurant partnerships.

• Use limited time offers and discounts to increase customer engagement and drive revenue.

4. Strengthen Underperforming Restaurant Categories

• Maintain the strong performance of Biryani and the steady performance of Indian cuisine.

• Improve Fast Food and Chinese categories through better restaurant options and promotions.

• Give greater attention to the Desserts and Café categories, which show the weakest performance.

• Introduce new, varied, and trending food items in these categories.

• Use discounts, combo offers, and targeted promotions to increase customer engagement and category sales.

5. Reduce Delivery Time and Improve Customer Satisfaction

Focus on reducing long delivery times, particularly orders taking around 40–50+ minutes.

Optimize delivery routes to avoid traffic prone and congested areas where possible.

Improve coordination between restaurants and delivery partners to minimize delays.

Aim for faster delivery times, ideally around 20–25 minutes where operationally feasible.

Faster and more reliable delivery can help improve customer satisfaction and ratings.

Conclusion

The Food Delivery Business Performance & Market Analytics project successfully analysed order data, revenue performance, customer satisfaction, delivery efficiency, city wise performance, and restaurant category trends using Microsoft Excel, Python, MySQL, and Power BI. The analysis identified Kochi as the best performing city, Biryani as the strongest restaurant category, and April as the peak revenue month. The dashboard also highlighted opportunities to improve customer satisfaction through faster delivery and strengthen underperforming restaurant categories. Overall, the project demonstrates how data analytics can transform food delivery data into meaningful business insights and support better operational planning, marketing strategies, customer satisfaction, and data driven business decisions.